Thomas Puskas
KECE304 Circuits 2 Lab Week 05 Lab Report
KUID: 2026962128

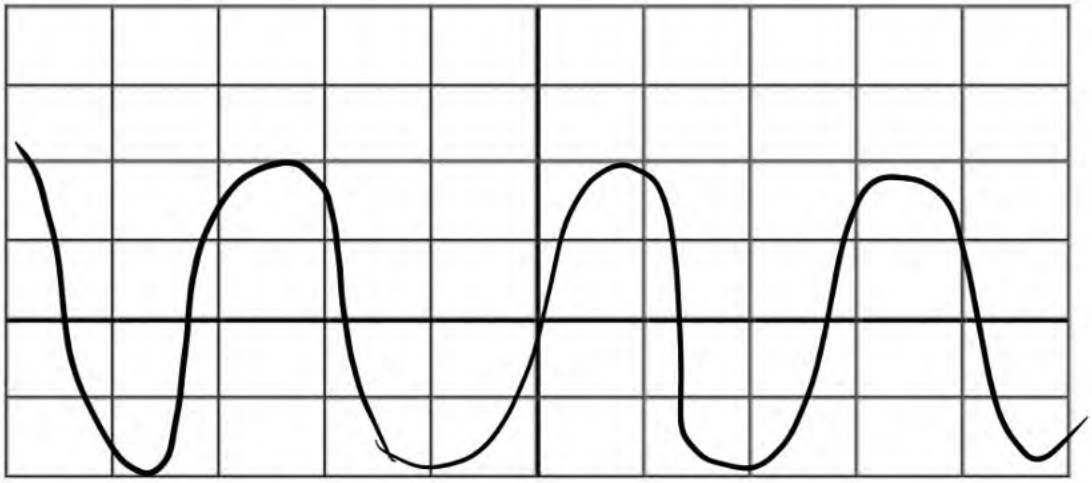
Experiment 3: Frequency Response

Purpose: We have looked at small signal frequencies and how they interact with a MOSFET. This experiment takes it further to see the frequency limits on each MOSFET and the parasitic capacitances in a MOSFET. By gradually increasing a frequency we can find the bandwidth (what frequencies the MOSFET operates at) and see how a MOSFET reacts different frequencies.

Materials: 1kΩ ResistorΩ x 1, 10kΩ ResistorΩ x 2, 10μF Capacitor x 1, 1nF Capacitor x 1, NMOS 2N7000 x 1

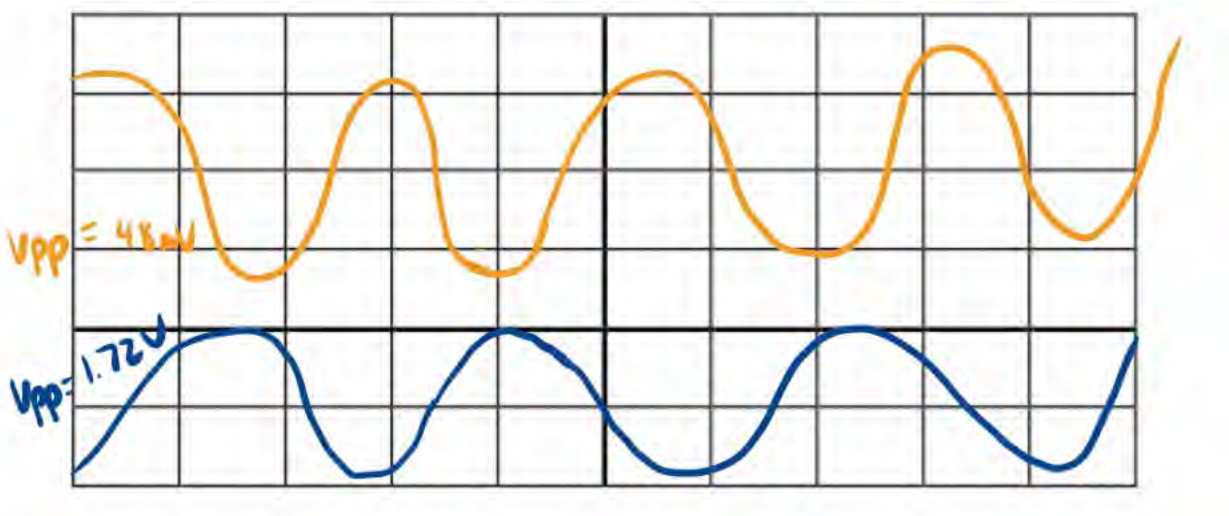
3.1 Biasing Circuit for input signal

3.



3.2. Frequency Response of a Common-Source Amplifier

2.




3.

| $V_{in}$ [Vpp] | $V_{out}$ [$V_{pp}$] | $\|Av\|$ $[V/V]$ |
|---|---|---|
| 0.0488 | 1.72 | 35.246 |

4.

| $gm$ [mA/A] | $\|Av\|$ $[V/V]$ |
|---|---|
| 18.8 | 18.8 |

5.

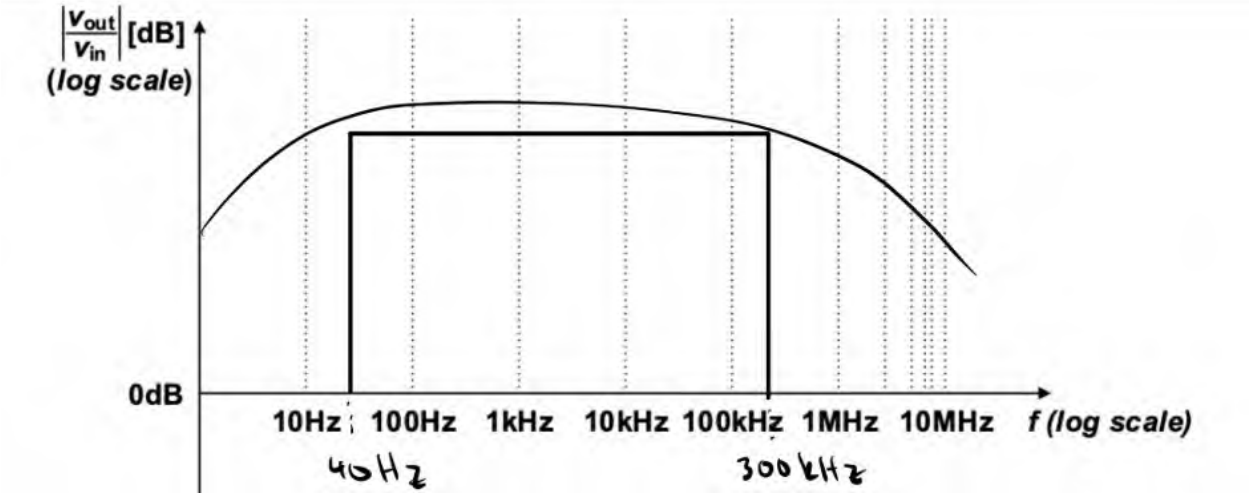
| Frequency | $V_{out}$ [$V_{pp}$] | $\|Av\|$ $[V/V]$ |
|---|---|---|
| 10 Hz | 0.968 | 16.13 |
| 100 Hz | 1.68 | 28 |
| 1 kHz | 1.76 | 29.33 |
| 10 kHz | 1.70 | 34 |
| 100 kHz | 1.64 | 27.33 |
| 1 MHz | 0.904 | 15.07 |
| 2 MHz | 0.528 | 10.56 |
| 3 MHz | 0.360 | 7.2 |
| 4 MHz | 0.272 | 5.44 |

| | | |
|---|---|---|
| 5 MHz | 0.224 | 3.73 |

6.

| $\|Av\|\ [V/V]$ | $f_{3-dB}[Hz]$ | 3dB bandwidth |
|---|---|---|
| 26.33 | 40 | 299.960k |
| | 300k | |

7.




8.

By using the formula given in the lab, we can find the pole frequency.

| $f_{p1}\ [Hz]$ | $f_{p2}\ [Hz]$ |
|---|---|
| 17.51 | 3.9 M |

3.3 Frequency Response Variation with a Output Load Capacitor

2.

| Frequency | $V_{out}$ [$V_{pp}$] | $\|Av\|\ [V/V]$ |
|---|---|---|
| 10 Hz | 0.864 | 14.4 |
| 100 Hz | 1.59 | 26.5 |
| 1 kHz | 1.59 | 26.5 |

| | | |
|---|---|---|
| 10 kHz | 1.54 | 25.6 |
| 100 kHz | 1.29 | 21.5 |
| 1 MHz | 0.238 | 3.97 |
| 2 MHz | 0.121 | 2.02 |
| 3 MHz | 0.082 | 1.37 |
| 4 MHz | 0.066 | 1.1 |
| 5 MHz | 0.044 | 0.73 |

3.

| $\|Av\|\ [V/V]$ | $f_{3-dB}[Hz]$ | 3dB bandwidth |
|---|---|---|
| 23.5 | 17.95 | 153.8K |
| | 153.77K | |

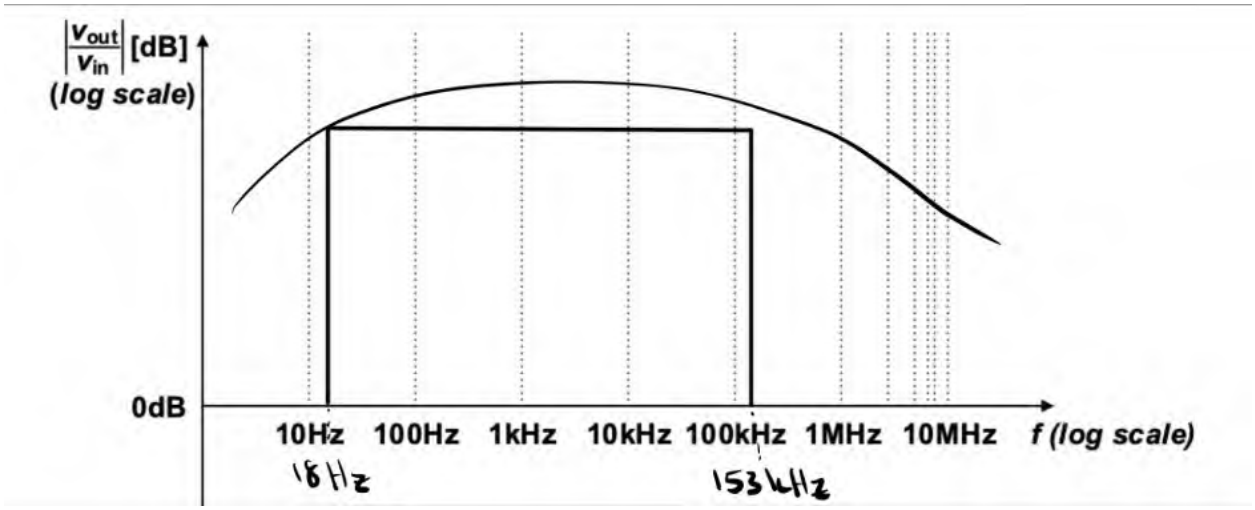
4.

| $f_u\ [kHz]$ |
|---|
| 4.1M |

6.

In this amplifier, we are using a 1 nF capacitor to simulate a large output capacitor. If we are to design an amplifier with high bandwidth (higher than 1 MHz), the transistor with a large output capacitor is **NOT** desirable.

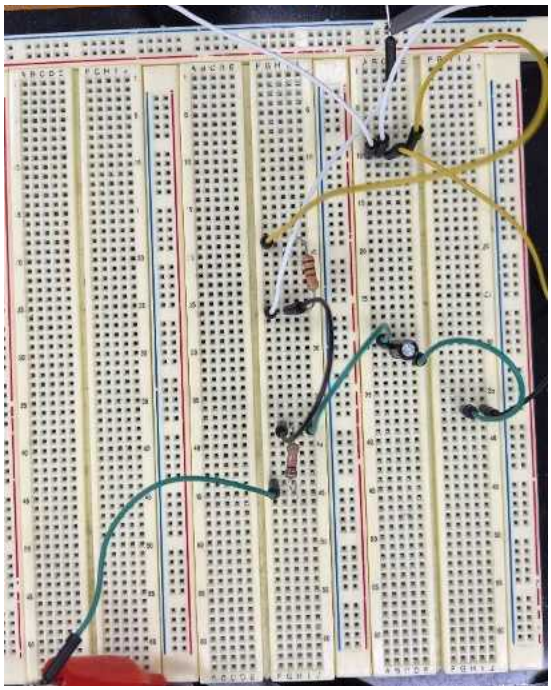
7.

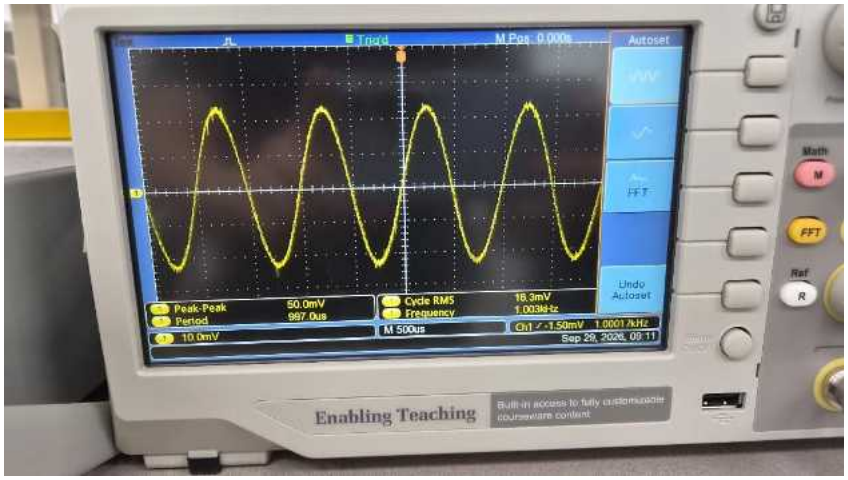


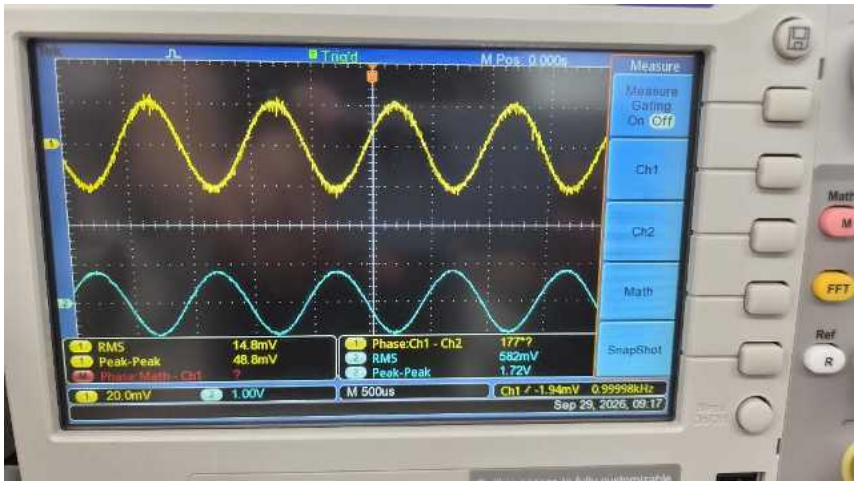

8.

By using the formula given in the lab, we can find the pole frequency.

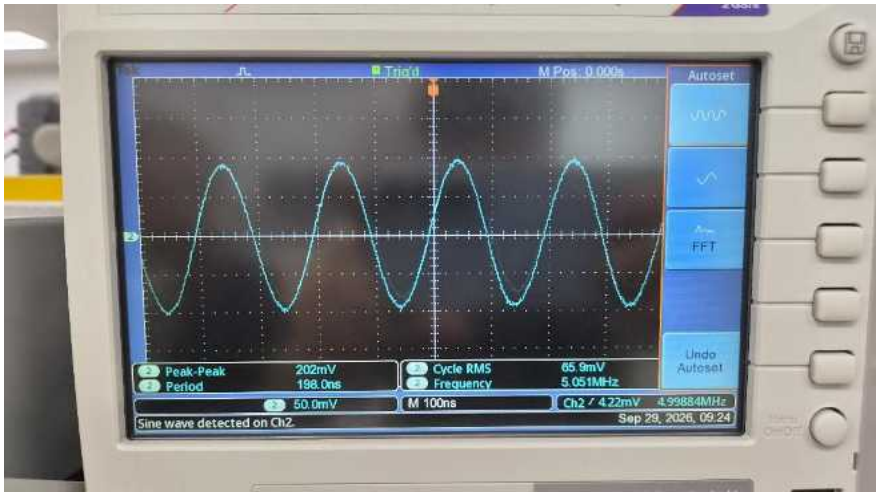
| $f_{p1}$ $[kHz]$ | $f_{p2}$ $[kHz]$ |
|---|---|
| 17.51 | 152.91k |

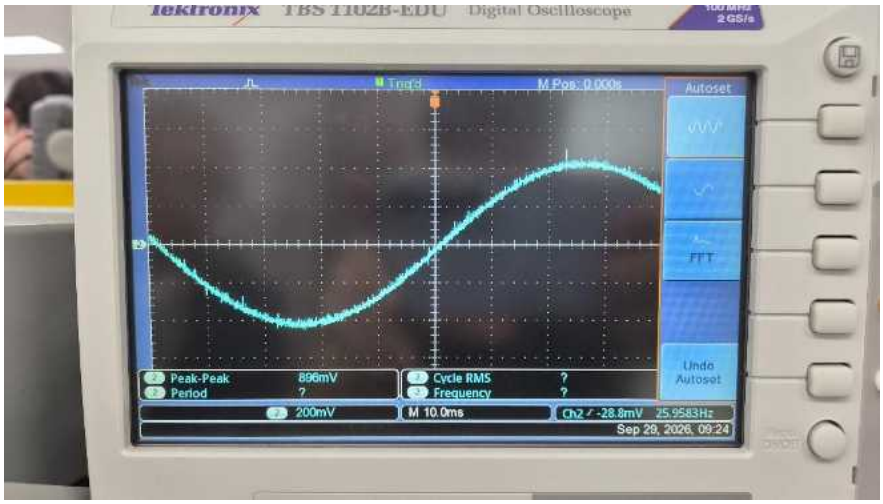


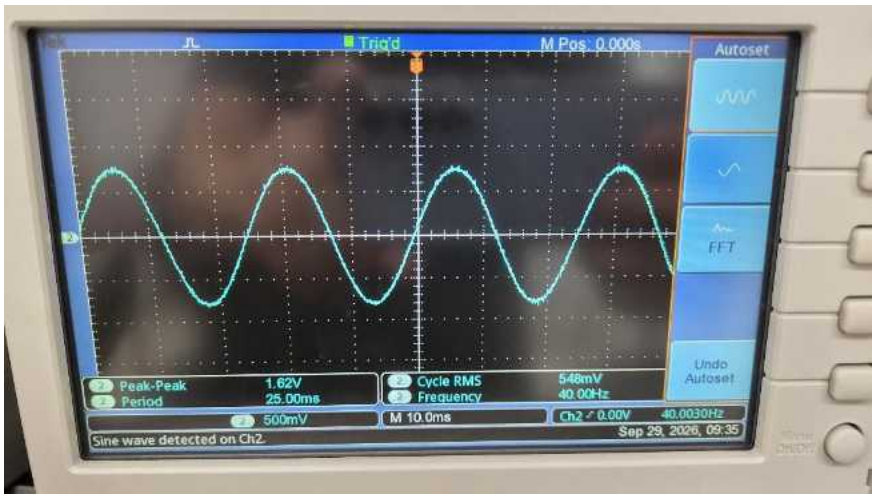

Trig'd
M Pos: 0.000s
Autoset
FFT
Undo Autoset
Peak-Peak 50.0mV
Period 987.0us
Cycle RMS 18.3mV
Frequency 1.003kHz
10.0mV
M 500us
Ch1 ✓ -1.50mV 1.0001kHz
Sep 28, 2026, 09:11
Enabling Teaching


Trig'd
M Pos: 0.000s
Measure
Measure Gating On Off
Ch1
Ch2
Math
SnapShot
RMS 14.8mV
Peak-Peak 48.8mV
Phase:Ch1 - Ch2 177°?
RMS 582mV
Peak-Peak 1.72V
20.0mV
1.00V
M 500us
Ch1 ✓ -1.94mV 0.99998kHz
Sep 29, 2026, 09:17


GW INSTEK AFG-2005 Arbitrary Function Generator
1.0000 kHz
FREQ
0.060 Vpp
AMPL
0.00 V
OFST
FUNC
FREQ
AMPL
OFST
DUTY
Point
Value
ARB
Save/Recall
Hz/Vpp


Trig'd
M Pos: 0.000s
Autoset
FFT
Undo
Autoset
Peak-Peak 202mV
Period 198.0ns
Cycle RMS 65.9mV
Frequency 5.051MHz
50.0mV
M 100ns
Ch2 ∕ 4.22mV 4.99884MHz
Sine wave detected on Ch2
Sep 29, 2026, 09:24

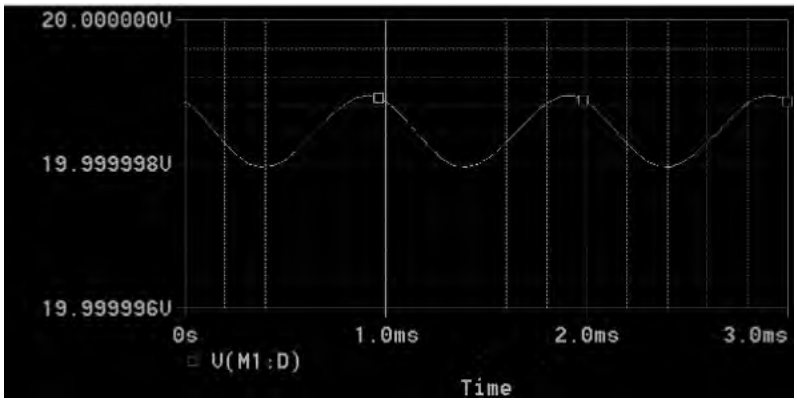

Tektronix TBS 1102B-EDU Digital Oscilloscope
2 GS/s
Trig'd
M Pos: 0.000s
Autoset
FFT
Undo
Autoset
Peak-Peak 896mV
Period ?
Cycle RMS ?
Frequency ?
200mV
M 10.0ms
Ch2 ∕ -28.8mV 25.9583Hz
Sep 29, 2026, 09:24


Trig'd
M Pos: 0.000s
Autoset
FFT
Undo
Autoset
Peak-Peak 1.62V
Period 25.00ms
Cycle RMS 548mV
Frequency 40.00Hz
500mV
M 10.0ms
Ch2 ∕ 0.00V 40.0030Hz
Sine wave detected on Ch2
Sep 29, 2026, 09:35

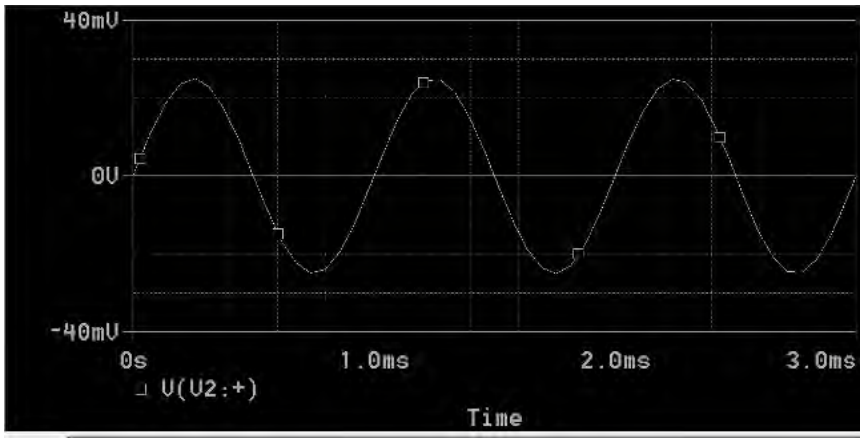
PSPICE:

3.2
2:

Vout




Vin:

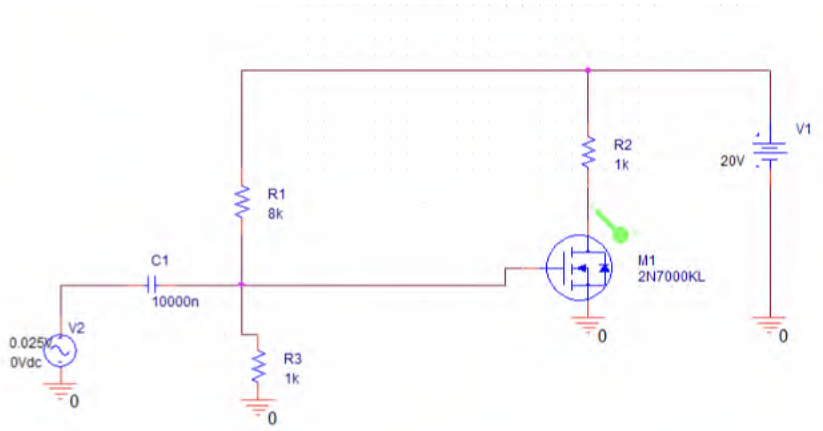



Pspice:

3.

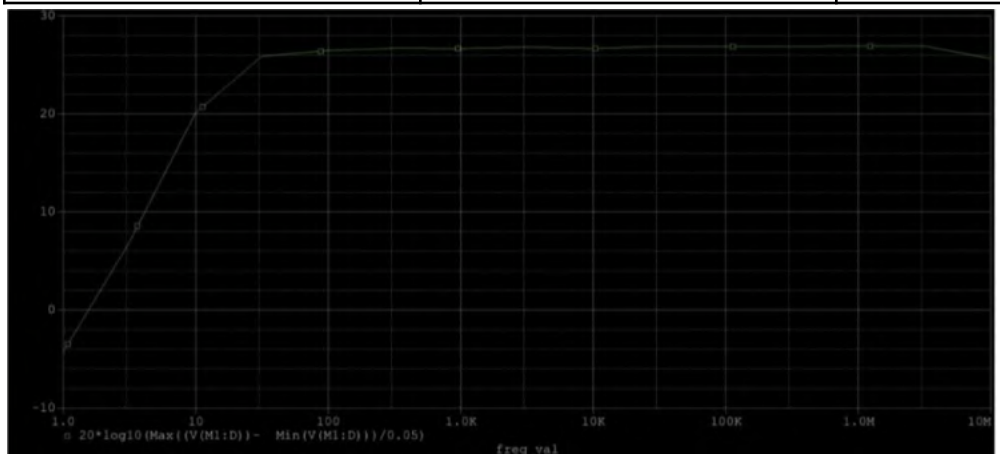
| $V_{in}$ [Vpp] | $V_{out}$ [$V_{pp}$] | $\|Av\|$ $[V/V]$ |
|---|---|---|
| 49.827 m | 1.0983 | 22.04 |

5.




6.

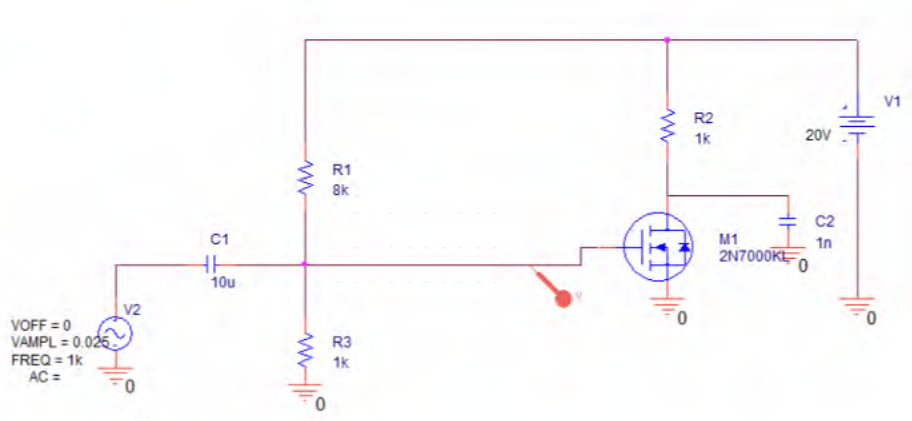
| $\|Av\|\ [V/V]$ | $f_{3-dB}[Hz]$ | 3dB bandwidth |
|---|---|---|
| 17.95 | 31 | 4.38 MHz |
| | 2.383 Mhz | |



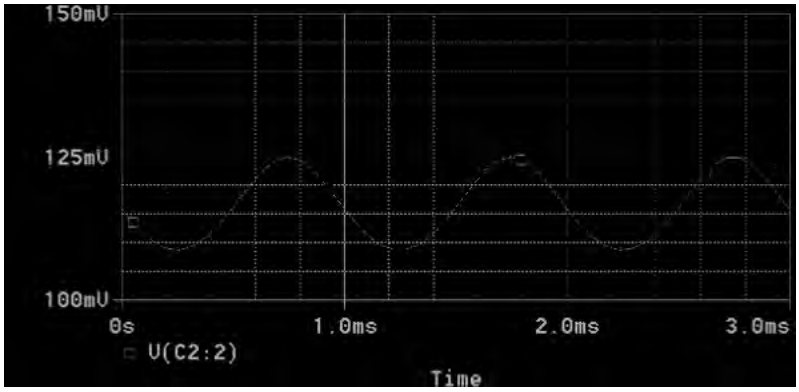

8.

| $f_{p1}\ [Hz]$ | $f_{p2}\ [Hz]$ |
|---|---|
| 17.90 | 4.4 M |

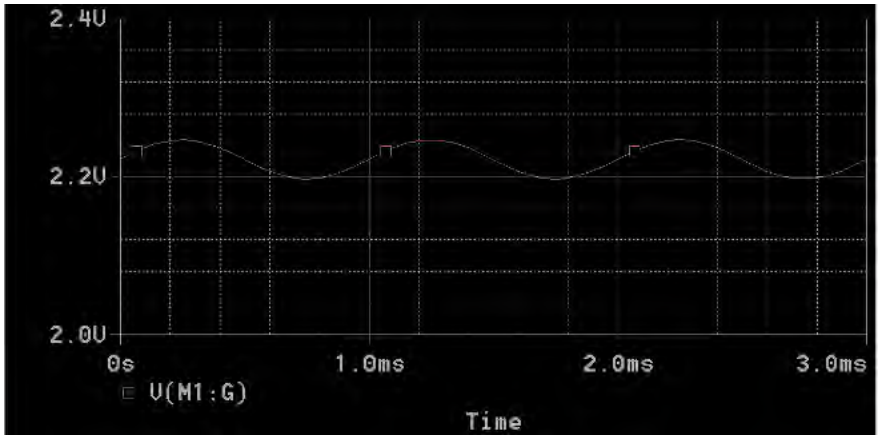
3.3

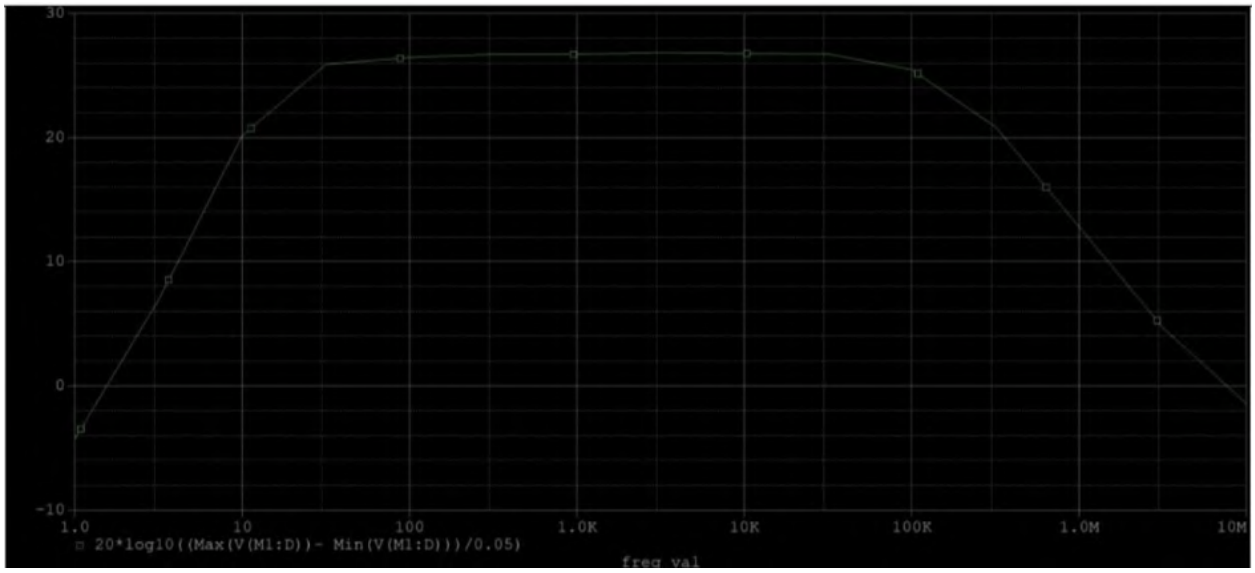

R2
1k
V1
20V
R1
8k
C1
10u
C2
1n
M1
2N7000KL
V2
VOFF = 0
VAMPL = 0.025
FREQ = 1k
AC =
R3
1k
0

Vout


150mV
125mV
100mV
0s
1.0ms
2.0ms
3.0ms
V(C2:2)
Time

Vin


2.4V
2.2V
2.0V
0s
1.0ms
2.0ms
3.0ms
V(M1:G)
Time




4. Comparison

Comparing Pspice to real life scenario we see that the numbers are relatively close but most definitely noy the same. There could be a multitude of reasons for this such as human error in the creation of our circuit or the transistors or NMOS could have slight differences due to manufacturing discrepancies compared to the ideal parts in PSPICE. There also could be some small deviations in the wave form generator to DC voltage source that also amplifies the differences. Although PSPICE is more ideal. It is important to do experiments in person and understand real world effects and how they impact data, but also realize that PSPICE is a great tool and useful for getting ideal data. Considerable differences could occur because a 8kOhm resistor was used in PSPICE and a 10kOhm Resistor was used in real life.